

Analysis and Modeling of Radio-Frequency Interference for Space Communication Links

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Abstract—The increasing density of satellite constellations and terrestrial wireless systems have elevated the risk of radio-frequency interference (RFI) to space communication links. This paper presents an ITU-R-aligned modeling framework for analyzing RFI across the S-, X-, Ku-, K-, and Ka-bands, with emphasis on quantifying RFI-induced attenuation of link margins.

Interference power is expressed as a function of frequency, geometry, and antenna discrimination, while other propagation effects and secondary impairments, such as atmospheric fading and pointing errors, are treated as fixed parameters to isolate the impact of RFI. The framework supports the analysis of aggregate interference scenarios using probabilistic metrics consistent with ITU-R Recommendations SA.609 and SA.1157. It also enables direct comparison across the three interference scenarios and representative distance cases such as GEO, Moon, and Mars.

Index Terms—Radio Frequency Interference, Satellite Communications, ITU-R, Spectrum Sharing, Cross-Band Modeling, Attenuation.

I. INTRODUCTION

Recent studies have shown that satellite systems operating in their assigned frequency bands are increasingly exposed to interference from terrestrial and other space-based transmitters, motivating detailed band-specific interference analyses such as those performed for the S- and Ku-bands in [1], [2]. At the same time, the deployment of large Non-Geostationary Satellite Orbit (NGSO) constellations, such as Starlink, has intensified aggregate interference scenarios affecting a wide range of space communication services [3], [4].

The International Telecommunication Union Radiocommunication Sector (ITU-R) establishes internationally recognized technical recommendations for spectrum allocation, interference assessment, and protection criteria for satellite services. These recommendations provide the regulatory and methodological foundation for evaluating compatibility between space and terrestrial systems.

Existing RFI studies are often limited to individual frequency bands or specific system configurations [5], [6]. While these works provide valuable insight, they do not offer a consistent basis for comparing how interference translates into link margin attenuation across multiple space communication bands. This paper addresses this gap by developing a modeling framework applicable to the S-, X-, Ku-, K-, and Ka-bands, enabling direct comparison under identical interference assumptions.

These bands exhibit distinct physical characteristics: higher-frequency bands always support larger bandwidths and higher achievable data rates, while lower-frequency bands offer greater robustness against atmospheric attenuation, rain fading, and pointing errors. A unified modeling framework therefore enables evaluation of how interference impacts systems with fundamentally different propagation and implementation trade-offs.

Table I summarizes the approximate frequency ranges of the satellite communication bands considered in this study.

TABLE I
SATELLITE COMMUNICATION BANDS CONSIDERED IN THIS STUDY

Band	Frequency Range
S-band	2 – 4 GHz
X-band	8 – 12 GHz
Ku-band	12 – 18 GHz
K-band	18 – 27 GHz
Ka-band	27 – 40 GHz

II. RELATED WORK

RFI in space communication systems has been widely studied, but much of the existing work focuses on specific frequency bands, services, or deployment scenarios. While these studies provide valuable insight into band-dependent interference mechanisms, differences in assumptions, modeling approaches, and performance metrics often make direct comparison across frequency bands difficult.

Broadly, prior research on RFI in satellite communications can be grouped into three main directions: identifying dominant interference sources, evaluating their impact on communication performance, and developing models to predict and quantify these effects. The following subsections summarize these contributions and highlight their relevance to the present work.

A. Dominant RFI Mechanisms Across Frequency Bands

The mechanisms that generate interference in satellite systems can be categorized both by frequency band and by their spectral relationship to the victim signal. In-band interference occurs when interfering signals fall within the same frequency allocation as the desired signal, directly degrading the received Signal-to-Noise Ratio (SNR). Adjacent-band interference arises from signals in neighbouring channels

due to imperfect filtering or spectral leakage, while out-of-band interference, including harmonic emissions, originates from unintended radiation outside the allocated band. These different interference types vary in their origin, spectral characteristics, and impact on receiver performance, and their effect depends strongly on receiver filtering, antenna characteristics, and system design.

The dominant sources and characteristics of interference also vary significantly across frequency bands due to differences in spectrum usage and propagation behavior. In lower-frequency bands such as the S-band, interference is often caused by coexistence with terrestrial wireless and mobile communication systems, with geographically distributed transmitters operating within or near the same allocations [1].

At higher frequencies, such as the Ku- and Ka-bands, interference scenarios often involve terrestrial microwave links and interactions between multiple satellite systems. In these densely deployed environments, NGSO constellations with many satellites can make uplink interference and cross-system interference more prominent [2]. These developments have increased the complexity of interference environments and highlight the need for modeling approaches capable of capturing both geometric effects and multi-source interference behavior [3], [4].

B. Band-Specific Interference Studies

A large portion of the literature focuses on detailed studies within individual frequency bands. In the S-band, coexistence between satellite and terrestrial systems has been widely studied, with interference levels strongly influenced by separation distance, antenna geometry, and deployment density. Reported results indicate that even moderate interference can lead to noticeable SNR degradation, particularly in mission-critical communication links [1].

For these high-frequency bands, interference has also been studied in the context of terrestrial microwave uplinks and high-throughput satellite systems. These studies emphasize the importance of antenna discrimination, spatial geometry, and network density in determining interference severity [2].

For higher-frequency bands such as Ku and Ka, propagation-focused studies also highlight the sensitivity of these bands to environmental effects such as rain attenuation [7]. This is due to their stronger sensitivity to atmospheric effects at higher frequencies. However, this work isolates the impact of RFI, so these secondary propagation effects are simplified in the present model. These differences motivate a modeling approach that can capture both geometric effects and multi-source interference behavior.

C. Quantifying the Impact of RFI on Communication Performance

The impact of interference on communication performance is typically evaluated using metrics such as the Carrier-to-Interference Ratio (C/I), SNR, and Bit Error Rate (BER) [5], [6]. These metrics provide a direct link between interference levels and communication reliability.

The ITU-R has established widely adopted frameworks for interference analysis in satellite systems. Recommendation ITU-R S.1325 defines methodologies for evaluating aggregate interference, while Recommendations SA.609 and SA.1157 specify protection criteria for space research services. These approaches are based on probabilistic metrics, where interference is characterized in terms of the fraction of time that performance thresholds are exceeded.

Such probabilistic formulations are used because interference conditions in satellite systems can change with geometry, transmitter activity, and operational setup. Time-based exceedance metrics therefore provide a practical way to assess system performance under realistic conditions.

D. Generic Versus Band-Specific Modeling Approaches

Existing RFI studies generally adopt either band-specific or unified modeling approaches. Band-specific studies focus on detailed modeling of individual frequency bands and system configurations, often incorporating highly specialized assumptions. While this can yield accurate results for a particular scenario, it limits the ability to compare interference behavior across bands.

In contrast, unified modeling frameworks provide a common structure that can be applied consistently across multiple frequency bands. By maintaining common assumptions and using standardized performance metrics, these approaches enable direct comparison of interference sensitivity under equivalent conditions.

The present work follows this second approach by developing a unified modeling framework for S-, X-, Ku-, K-, and Ka-bands. By maintaining consistent assumptions and isolating RFI effects, the model enables direct comparison of how different frequency bands respond under equivalent conditions.

The next section introduces the modeling framework used in this study and describes the assumptions and formulations underlying the analysis.

III. GENERIC RFI MODEL

A. Scope and Assumptions

As presented above, the model is designed to isolate and evaluate the impact of RFI on link performance under controlled conditions. Secondary impairments such as atmospheric attenuation, polarization mismatch, and hardware-related losses are included in the formulation for completeness, but are held fixed in the simulations; in particular, atmospheric attenuation is set to zero in the numerical evaluation, i.e., $L_{\text{atm}} = 0$.

A deterministic pointing margin is incorporated to represent practical antenna misalignment. This margin is defined as a fixed fraction of the antenna Half-Power Beamwidth (HPBW) and is added to the nominal off-axis angle used in the antenna gain evaluation. In the simulations, the pointing margin is set to

$$\theta_{\text{pm}} = \alpha \theta_{3\text{dB}}, \quad \alpha = 0.3, \quad (1)$$

TABLE II
COMPUTED HPBW AND POINTING MARGIN FOR EACH FREQUENCY BAND

Band	f (GHz)	$\theta_{3\text{dB}}$ (deg)	θ_{pm} (deg)
S-band	3.0	5.83	1.75
X-band	8.4	2.08	0.62
Ku-band	14.0	1.25	0.37
K-band	22.0	0.79	0.24
Ka-band	32.0	0.55	0.16

The antenna HPBW is approximated as

$$\theta_{3\text{dB}} \approx \frac{70 \lambda}{D}, \quad (2)$$

where λ is the wavelength and D is the antenna diameter. This expression is used directly in the numerical implementation to compute both the beamwidth and the associated pointing margin.

To illustrate the resulting frequency dependence, Table II lists the HPBW and pointing margin computed from (2) at representative center frequencies for each band, using the assumed antenna diameter $D = 1.2$ m and the pointing-margin factor $\theta_{\text{pm}} = 0.3 \theta_{3\text{dB}}$.

In the analytical formulation, θ_i denotes the nominal off-axis angle, while in the simulations it is replaced by an effective angle that incorporates the pointing margin, as described below.

B. Interference Model

The antenna characteristics described above directly influence the level of interference received at the victim terminal through the off-axis gain $G_{\text{rx}}(f, \theta)$. In particular, the frequency-dependent beamwidth and the associated pointing margin determine the effective angular separation between the interferer and the main beam, and therefore the degree of interference suppression achieved by the antenna pattern.

Based on these considerations, the received interference power from a single source i is modeled as

$$I_i = EIRP_i(f) - L_{\text{fs}}(f, d_i) - L_{\text{atm}}(f, d_i, W_i) + G_{\text{rx}}(f, \theta_i) - L_{\text{misc}} \quad (3)$$

where f is the operating frequency, d_i is the interferer distance, θ_i is the nominal off-axis angle at the victim antenna (later replaced by the effective angle θ_i^{eff} in the implemented model), and W_i denotes the propagation state. $EIRP_i$ is the equivalent isotropically radiated power of the interferer, L_{fs} is the free-space path loss [8], L_{atm} (which is set to 0), G_{rx} is the receive antenna gain evaluated using an ITU-R-based antenna pattern model [9], [10], and L_{misc} accounts for fixed implementation losses.

In the implemented simulations, the effective off-axis angle is modified to include the deterministic pointing margin:

$$\theta_i^{\text{eff}} = \theta_i + \theta_{\text{pm}}, \quad (4)$$

with $\theta_{\text{pm}} = 0.3 \theta_{3\text{dB}}$. The receive antenna gain G_{rx} is evaluated using this effective angle together with the ITU-R S.1528 radiation pattern envelope, introducing an additional frequency-dependent suppression of off-axis interference.

When multiple interferers are present, their contributions are combined in the linear power domain [11]:

$$I_{\text{agg}} = 10 \log_{10} \left(\sum_i 10^{I_i/10} \right). \quad (5)$$

C. Carrier, Noise, and Attenuation

The impact of interference on link performance is quantified through standard link-budget metrics. The received carrier power is expressed as

$$C = EIRP_{\text{tx}} + G_{\text{rx}} - L_{\text{path}} - L_{\text{other}}, \quad (6)$$

where $EIRP_{\text{tx}}$ is the transmit equivalent isotropically radiated power, G_{rx} is the receive antenna gain in the direction of the desired signal, L_{path} is the propagation loss, and L_{other} represents fixed implementation losses.

The thermal noise power in bandwidth B is given by

$$N = 10 \log_{10}(k_B T_{\text{sys}} B), \quad (7)$$

where k_B is Boltzmann's constant, T_{sys} is the system noise temperature, and B is the receiver bandwidth.

The interference-free signal-to-noise ratio (SNR) is

$$\text{SNR}_0 = C - N, \quad (8)$$

while in the presence of aggregate interference it becomes [11]

$$\text{SNR}_I = 10 \log_{10} \left(\frac{10^{C/10}}{10^{N/10} + 10^{I_{\text{agg}}/10}} \right). \quad (9)$$

The degradation of the link margin due to interference is then defined as

$$\Delta \text{SNR} = \text{SNR}_0 - \text{SNR}_I. \quad (10)$$

D. Probabilistic Evaluation

In practical satellite communication scenarios, interference is inherently time-varying due to changes in transmitter activity and system dynamics. To capture this behaviour, the aggregate interference power is modeled as a stochastic process by applying log-normal variability in the decibel domain. This produces a time series of interference levels from which the corresponding SNR degradation is evaluated.

Rather than relying solely on deterministic metrics, system performance is characterized using time-fraction exceedance probabilities:

$$P[\Delta \text{SNR}(t) > \Delta], \quad (11)$$

which quantify the fraction of time during which the SNR degradation exceeds a specified threshold Δ . In this work, thresholds of 1 dB and 3 dB are considered, consistent with the statistical post-processing approach described in ITU-R Recommendation S.1325 [11].

These probabilistic metrics provide a compact and physically meaningful description of interference impact, enabling

consistent comparison across frequency bands and interference scenarios. The formulation follows the statistical approach adopted in ITU-R Recommendations SA.609 and SA.1157 [12], [13].

The above formulation provides the analytical basis for evaluating interference effects under controlled conditions. The following section describes the simulation framework used to implement and analyze this model.

IV. SIMULATION SETUP

The generic RFI model described in Section III is evaluated using a Python script, in which analytical expressions derived from ITU-R recommendations are implemented and evaluated under defined scenario parameters. The resulting data are analyzed and visualized using a Jupyter notebook environment, ensuring a consistent and reproducible workflow across all simulated scenarios.

Simulations are performed for the S-, X-, Ku-, K-, and Ka-bands. For each band b , the victim link is characterized by the operating frequency f_b , transmit equivalent isotropically radiated power $EIRP_{tx}$, antenna diameter D , antenna efficiency η , system noise temperature T_{sys} , and receiver bandwidth B . In the numerical evaluation, these parameters are held constant across all bands, with $EIRP_{tx} = 40$ dBW, $D = 1.2$ m, $\eta = 0.65$, $T_{sys} = 300$ K, and $B = 1$ MHz. As a result, frequency f_b is the primary variable distinguishing each band, allowing direct comparison of interference effects under otherwise identical system conditions.

The victim receiver is assumed to operate in a geostationary orbit with a propagation distance of $d = 36,000$ km. Receive antenna gain and beamwidth are derived from the antenna geometry using the HPBW expression in (2), while off-axis discrimination is modeled using the ITU-R S.1528 antenna envelope [9]. A deterministic pointing margin equal to 30% of the antenna HPBW is applied, consistent with the model formulation, and incorporated into the effective off-axis angle used in the interference calculations.

Interference power from each transmitter is computed using the equation defined in (3). While the framework supports multiple simultaneous interferers, the present analysis considers a single interferer in order to isolate fundamental interference behaviour. When multiple interferers are included, their contributions are combined using the linear-domain aggregation described in (5). Carrier power, thermal noise power, and the resulting SNR values are evaluated using the relations in (6)–(10).

The simulations consider three interference levels—weak, moderate, and strong—defined by different interferer transmit powers, while the remaining parameters are kept fixed, as summarized in Table III.

To capture time-varying interference conditions, the aggregate interference power is modeled as a log-normal random process in the decibel domain. Random interference samples are generated using a standard deviation $\sigma = 6$ dB and a duty cycle of unity, producing a time series of interference levels

TABLE III
PARAMETERS DEFINING THE SIMULATED INTERFERENCE SCENARIOS

Parameter	Weak	Moderate	Strong
Interferer $EIRP_i$ (dBW)	10	20	30
Separation distance d_i (km)	36,000	36,000	36,000
Nominal off-axis angle θ_i (deg)	0.5	0.5	0.5
Log-normal std. dev. σ (dB)	6	6	6
Duty cycle	1.0	1.0	1.0

from which the corresponding SNR degradation is evaluated using the probabilistic formulation in (11).

Frequency-dependent behaviour is examined through discrete frequency sweeps within each band. The S-band is evaluated at representative edge frequencies (2 GHz and 4 GHz), while higher-frequency bands are sampled at multiple discrete points across their operating ranges (e.g., 8–10 GHz for X-band, 12–16 GHz for Ku-band, 18–24 GHz for K-band, and 26–40 GHz for Ka-band).

The simulations report several performance metrics derived from the link budget and interference calculations, including the baseline SNR (SNR_0), aggregate interference power (I_{agg}), C/I, SNR in the presence of interference (SNR_I), and the resulting interference-induced attenuation of the link margin (ΔSNR). To characterize temporal behaviour, exceedance probabilities are computed for attenuation thresholds of 1 dB and 3 dB using the time-fraction formulation in (11).

V. RESULTS

This section presents the numerical results obtained from the simulation framework described in Section IV. The analysis proceeds in two stages. First, the baseline performance of the communication link is evaluated to establish the interference-free reference SNR (SNR_0). Second, the impact of RFI is examined under three representative interference scenarios—weak, moderate, and strong—while also considering frequency variations within each band.

A. Baseline Link Performance and Frequency Dependence

Fig. 1 shows the baseline SNR when frequency is swept across the S-, X-, Ku-, K-, and Ka-bands. The baseline SNR remains nearly constant across the full frequency range.

This happens because the increase in antenna gain is balanced by the increase in path loss. As a result, the received carrier power stays almost unchanged, and so does the baseline SNR.

B. Impact of Interference Across Frequency Bands

The three interference levels are defined in Table III. Under weak interference, the SNR drop is very small across all bands.

Under moderate interference, the SNR loss becomes more visible, especially at lower frequencies.

Under strong interference, the SNR degradation is largest at the lower-frequency bands, while the higher-frequency bands are less affected.

Across all three cases, the same pattern appears: SNR degradation decreases as frequency increases. This is mainly

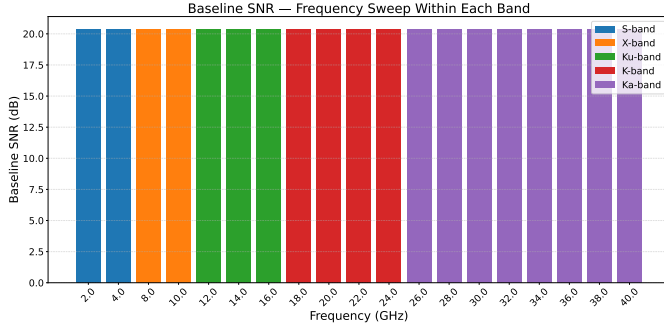


Fig. 1. Baseline SNR obtained when sweeping frequency across the range of each band.

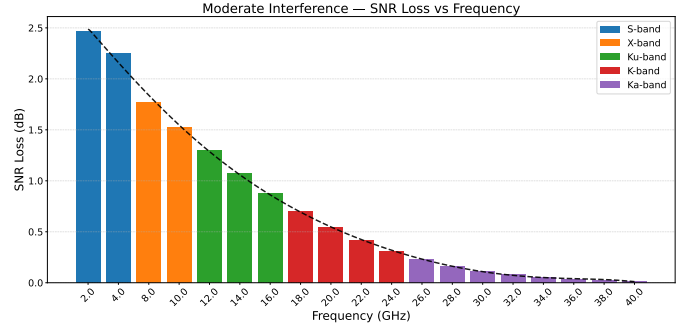


Fig. 3. RFI-induced SNR attenuation across frequency bands under moderate interference conditions.

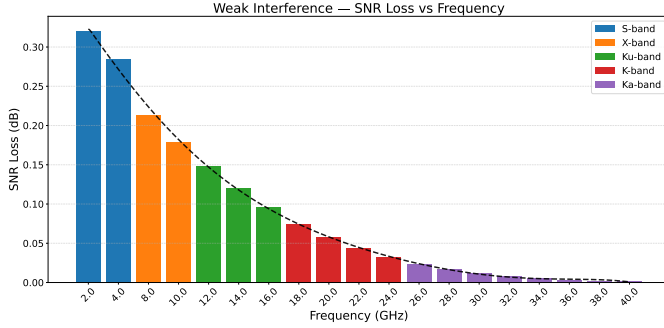


Fig. 2. RFI-induced SNR attenuation across frequency bands under weak interference conditions.

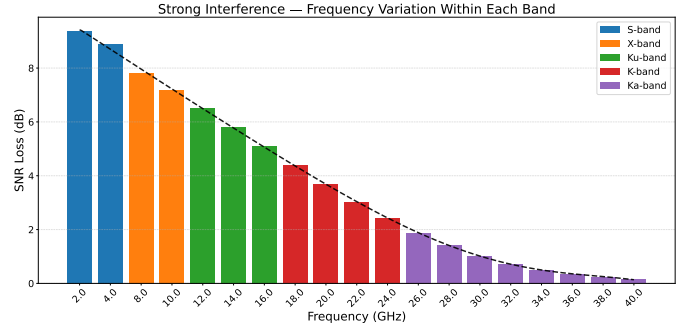


Fig. 4. RFI-induced SNR attenuation across frequency bands under strong interference conditions.

because higher-frequency bands use narrower beams, which reduce off-axis interference.

The frequency-dependent pointing margin reinforces this effect, since it also becomes smaller at higher frequencies.

C. Distance-Dependent Interference Behaviour

Propagation distance plays a critical role in determining interference impact. Fig. 5 shows the variation of SNR loss as a function of both frequency and link distance on a logarithmic colour scale.

The results demonstrate that interference impact decreases rapidly with increasing distance due to the growth of FSPL. At shorter distances, strong gradients in SNR loss are observed, indicating high sensitivity to interference. As distance increases, interference effects diminish significantly, and SNR degradation becomes negligible across all frequencies.

This behaviour highlights that, beyond antenna-related effects, propagation loss alone can strongly limit interference coupling, particularly in long-distance communication scenarios.

D. Interplanetary Scenario: Moon vs Mars

Figs. 6 and 7 show SNR degradation across frequency for lunar and Martian communication scenarios under the three interference levels defined in Table III. Figs. 8 and 9 illustrate the corresponding distance-dependent behavior through SNR loss heatmaps.

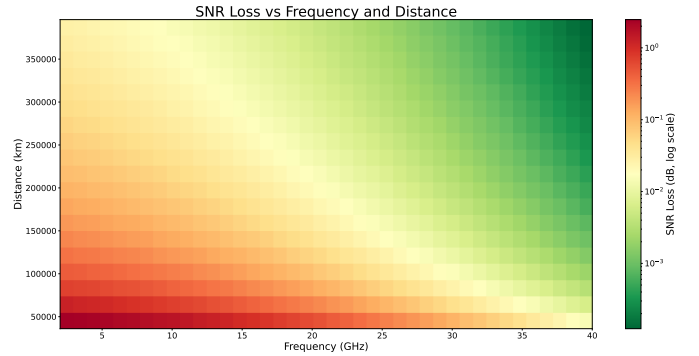


Fig. 5. SNR loss as a function of frequency and distance

The Moon scenario (approximately 3.84×10^5 km) exhibits noticeable SNR degradation, particularly under strong interference conditions where lower frequencies show losses exceeding 1 dB. In contrast, the Mars scenario (approximately 5.6×10^7 km) shows negligible SNR loss across all frequencies and interference levels due to extreme FSPL.

These results highlight the dominant role of propagation distance in suppressing interference. At lunar distances, antenna directivity provides moderate protection, but at Martian distances, path loss alone reduces interference to insignificant levels, effectively restoring noise-limited system behavior even under strong interference conditions.

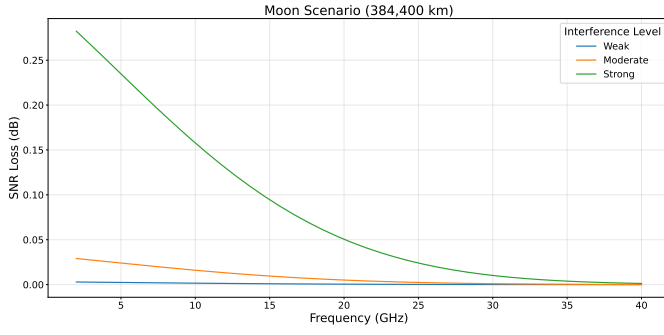


Fig. 6. SNR loss vs frequency for Moon scenario (384,400 km) under weak, moderate, and strong interference levels.

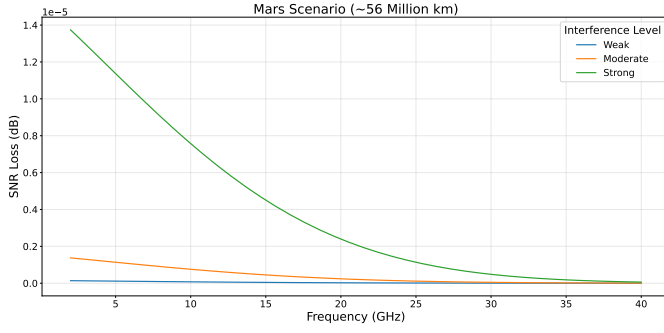


Fig. 7. SNR loss vs frequency for Mars scenario (~56 million km) under weak, moderate, and strong interference levels.

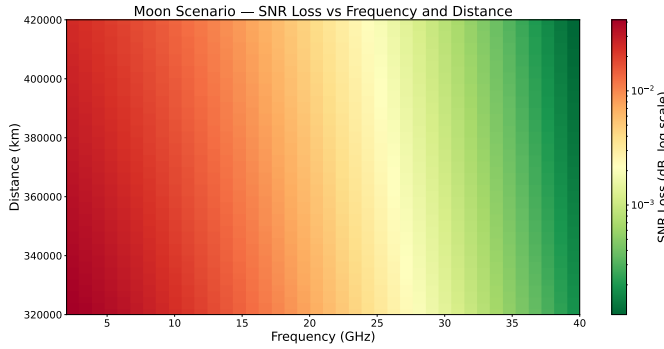


Fig. 8. SNR loss heatmap for Moon distances (320,000–420,000 km) vs frequency under moderate interference.

VI. DISCUSSION

The results show that RFI affects lower-frequency bands more strongly than higher-frequency ones. This is mainly due to antenna directivity: for a fixed antenna diameter, higher frequency leads to a narrower beamwidth and less off-axis interference. This is illustrated in Fig. 10.

The pointing margin also helps reduce interference. Since it is defined as a fraction of the beamwidth, it becomes smaller at higher frequencies. This means the interferer is effectively placed farther from the main beam, which further lowers the received interference.

As shown in the previous figures, the SNR degradation follows an approximately exponential decrease with frequency

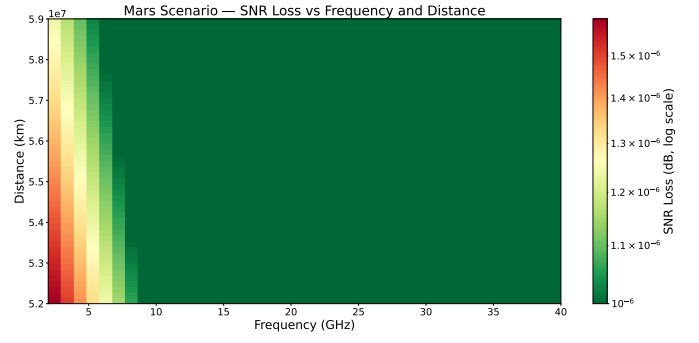


Fig. 9. SNR loss heatmap for Mars distances (52–59 million km) vs frequency under moderate interference.

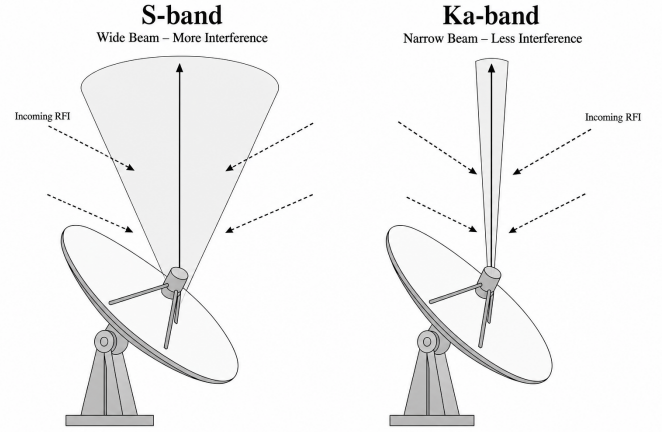


Fig. 10. Conceptual illustration of off-axis interference rejection for a lower-frequency band (S-band) and a higher-frequency band (Ka-band). The narrower beam at higher frequencies provides stronger rejection of off-axis RFI.

for all three interference scenarios. From the weak-interference plot, the fitted curve is:

$$\Delta\text{SNR}_{\text{weak}}(f) = 0.336 e^{-0.218f} \quad (12)$$

Similarly, from the moderate-interference plot, the fitted expression is:

$$\Delta\text{SNR}_{\text{moderate}}(f) = 2.627 e^{-0.196f} \quad (13)$$

For the strong-interference case, the fitted curve obtained from the corresponding figure is:

$$\Delta\text{SNR}_{\text{strong}}(f) = 10.370 e^{-0.143f} \quad (14)$$

These three cases can be summarized by the general exponential form

$$\Delta\text{SNR}(f) \approx a e^{-bf} \quad (15)$$

where the coefficients a and b depend on the interference level.

To relate this to the model, we start from the baseline SNR in (8), where $\text{SNR}_0 = C - N$. When interference is present,

the SNR is given by (9). Using the definition of link margin degradation in (10), we obtain

$$\begin{aligned}\Delta\text{SNR} &= \text{SNR}_0 - \text{SNR}_I \\ &= (C - N) - 10 \log_{10} \left(\frac{10^{C/10}}{10^{N/10} + 10^{I_{\text{agg}}/10}} \right) \\ &= 10 \log_{10} \left(\frac{10^{N/10} + 10^{I_{\text{agg}}/10}}{10^{N/10}} \right) \\ &= 10 \log_{10} \left(1 + 10^{\frac{I_{\text{agg}} - N}{10}} \right)\end{aligned}\quad (16)$$

If we define

$$X(f) = I_{\text{agg}}(f) - N, \quad (17)$$

then (16) becomes

$$\Delta\text{SNR}(f) = 10 \log_{10} \left(1 + 10^{\frac{X(f)}{10}} \right) \quad (18)$$

From the interference model in (3), the received interference from a single source is determined by EIRP, FSPL, atmospheric loss, antenna gain, and miscellaneous losses. For the simulations, the effective off-axis angle includes the pointing margin, and the aggregate interference is formed according to (5). This means that $I_{\text{agg}}(f)$ captures the combined effect of frequency-dependent path loss and antenna discrimination.

Using the same link-budget terms, the aggregate interference power can be written as

$$\begin{aligned}I_{\text{agg}}(f) &= \text{EIRP}_i - 32.45 - 20 \log_{10}(f_{\text{MHz}}) \\ &\quad - 20 \log_{10}(d_{i,\text{km}}) + G_{\text{max}} - 12 \left(\frac{\theta_{\text{eff}}}{\theta_{3\text{dB}}(f)} \right)^2,\end{aligned}\quad (19)$$

which shows explicitly that the interference term depends on both frequency and propagation distance. The path loss increases with frequency and distance, while the antenna gain term reduces the received interference at higher frequencies.

Substituting (19) into (17), we get

$$X(f) = I_{\text{agg}}(f) - N, \quad (20)$$

so the SNR degradation in (18) depends on both the interference level and the noise floor.

The exponential fit is only a compact way to describe the trend seen in the results. The results also show a clear change across the three interference levels. Under weak interference, the SNR degradation is very small. As the interference level increases, the degradation becomes more noticeable, especially at lower frequencies.

Distance has a similar effect. As the distance increases, path loss reduces the interference power. That is why the SNR degradation is much smaller at long distances. In the Mars scenario, the interference is almost negligible because the path loss is very large.

In short, interference sensitivity depends mainly on antenna directivity and propagation loss. Higher-frequency systems are better protected because of their narrower beams, and larger distances reduce interference through path loss. Together, these effects explain the performance trends seen in the simulated satellite communication links.

VII. CONCLUSION

This work proposes an ITU-R-aligned framework for studying radio-frequency interference in satellite communication links across the S-, X-, Ku-, K-, and Ka-bands. The model relates SNR degradation to the main link-budget factors, including interferer power, propagation loss, antenna characteristics, and noise.

The analytical formulation shows that SNR degradation depends on the interference-to-noise ratio, which is influenced by both frequency and propagation distance through FSPL and antenna directivity. This provides a clear physical interpretation of how interference affects link performance under different conditions.

Simulation results show that SNR degradation decreases smoothly with frequency, which can be approximated by an exponential function of the form:

$$\Delta\text{SNR}(f) \approx a e^{-bf} \quad (21)$$

The exponential fit is a compact way to summarize the trend seen in the results, and it should be interpreted as a descriptive fit rather than a physical law.

In summary, interference sensitivity is shaped by the combined effects of antenna directivity and propagation loss. Higher-frequency systems are better protected due to narrower beamwidths, while increased distance further reduces interference through path loss. These findings provide a consistent and physically grounded understanding of how interference behaves across different satellite communication frequency bands.

REFERENCES

- [1] L. Zhang, Z. Li, and S. Yang, "Evaluating the influence of satellite systems on terrestrial networks: Analyzing S-band interference," in *Proceedings of the 5th International Conference on Sensors and Information Technology*, Nanjing, China, 2025, pp. 834–839. DOI: 10.1109/ICSIT64877.2025.11009924.
- [2] A. Karaki, H. Abdalla, and H. Issa, "Investigating Ku-band up-link interference from terrestrial microwaves on satellite reception," in *Proceedings of the IEEE Middle East and North Africa Communications Conference (MENACOMM)*, Byblos, Lebanon, 2025, pp. 1–6. DOI: 10.1109/MENACOMM62946.2025.10911012.
- [3] E. Polo, L. Emiliani, and L. Luini, "Investigating NGSO constellation-induced interference on GSO system ground stations," in *Proceedings of the ESA Telecommunications and Navigation Conference (TTC)*, 2025.
- [4] A. Bouleux, "Interference induced by NGSO constellations on GSO systems," M.S. thesis, Politecnico di Milano, 2022.
- [5] S. Manshadi et al., "Protection of deep space missions from RFI caused by EESS satellites," in *URSI Radio Frequency Interference Workshop*, 2019.

- [6] A. Coppotelli et al., "Evaluating the risk of RFI between lunar and lagrangian space missions sharing the SRS bands," in *Proceedings of the ESA Telecommunications and Navigation Conference (TTC)*, 2025.
- [7] International Telecommunication Union, "Propagation data and prediction methods required for the design of earth-space telecommunication systems," ITU-R, Recommendation ITU-R P.618-13, 2017.
- [8] International Telecommunication Union, "Calculation of free-space attenuation," ITU-R, Recommendation ITU-R P.525, 2019.
- [9] International Telecommunication Union, "Satellite antenna radiation pattern for use as a design objective in the fixed-satellite service," ITU-R, Recommendation ITU-R S.1528, 2001.
- [10] International Telecommunication Union, "Mathematical model of average and related radiation patterns for line-of-sight point-to-point fixed wireless system antennas for use in coordination studies and interference assessment," ITU-R, Recommendation ITU-R F.1245, 2019.
- [11] International Telecommunication Union, "Simulation methodologies for interference in satellite systems," ITU-R, Recommendation ITU-R S.1325-3, 2015.
- [12] International Telecommunication Union, "Protection criteria for the space research service (near-earth)," ITU-R, Recommendation ITU-R SA.609-2, 2013.
- [13] International Telecommunication Union, "Protection criteria for the space research service (deep space)," ITU-R, Recommendation ITU-R SA.1157-5, 2018.
- [14] International Telecommunication Union, "Characteristics and protection criteria for receiving GNSS and aeronautical receivers operating in the RNSS bands," ITU-R, Recommendation ITU-R M.1903-1, 2023.
- [15] "Radio interference from satellites is threatening astronomy – a proposed zone for testing new technologies could head off the problem." Accessed: 22-Nov-2025, National Science Foundation. [Online]. Available: <https://par.nsf.gov/servlets/purl/10428373>.
- [16] International Telecommunication Union, "Recommendation itu-r f.1249-5: Technical and operational requirements for the automated spectrum management system," ITU-R, Tech. Rep., 2018.